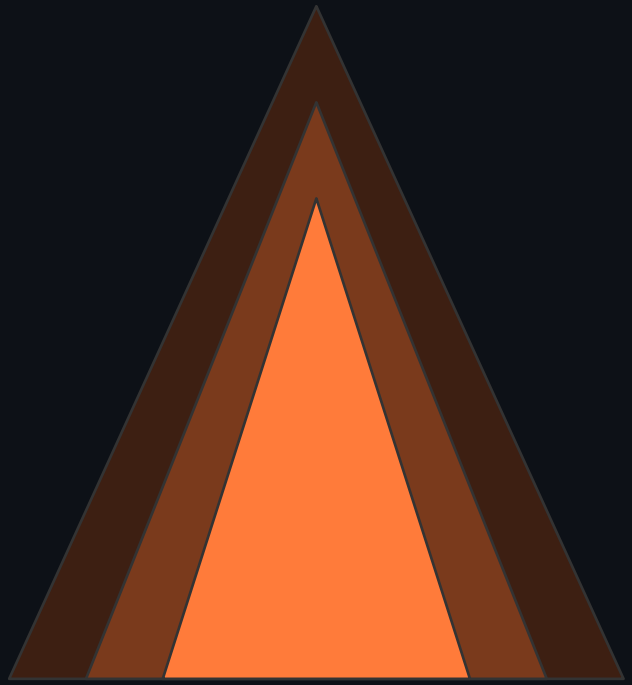
CNCF · Incubating

Volcano

Kubernetes done right for batch

Spark · Flink · Trino · ML on K8s



volcano.sh

What

Kubernetes was built for long-running services.
Batch is a different beast.

Volcano is the scheduler that fills the gap.

01
Gang scheduling
All-or-nothing pod groups

02
Queues & fair-share
Priority, preemption, hierarchy

03
Real Job CRD
Native batch semantics

04
Plugin scheduler
gang · drf · proportion · binpack

VOLCANO

How it works

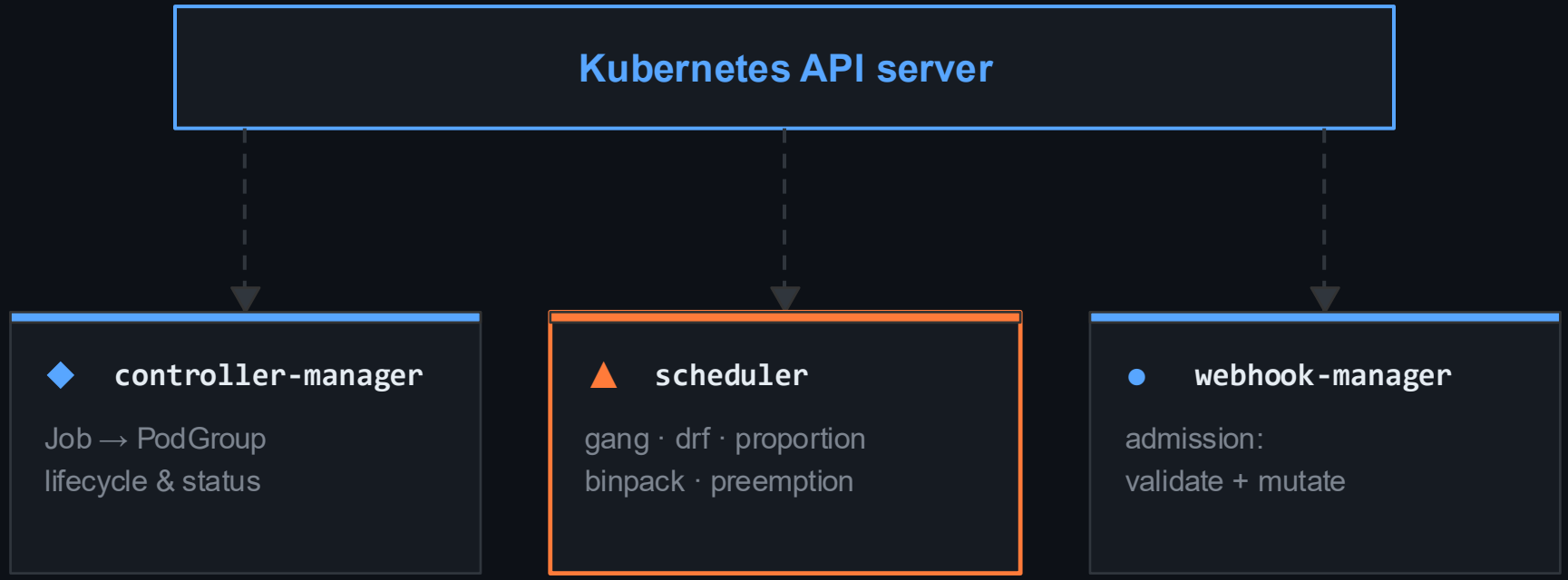
vcjob.yaml

```yaml
apiVersion: batch.volcano.sh/v1alpha1
kind: Job
metadata:
  name: quarterly-revenue-rebuild
spec:
  minAvailable: 6       # all 6 executors up together, or none
  schedulerName: volcano
  queue: gold-tier
  tasks:
    - replicas: 6
      name: spark-executor
      image: spark:3.5
```

Job → **PodGroup** → all executors scheduled together, or none.

VOLCANO

Internal architecture



CRDs: Job · Queue · PodGroup

Trade-offs

+ PROS

- Gang scheduling — no Spark/Flink deadlocks
- Real queues, fair-share, preemption
- Pluggable scheduler
- Native: Spark · Flink · Ray · PyTorch · MPI
- Production-grade

− CONS

- Extra CRDs, extra scheduler
- Friction with cluster-autoscaler
- Smaller community
- Overkill for stateless services
- HPA does not apply to vcjobs

Worth it if you run Spark · Flink · Ray · ML on K8s. *Otherwise no.*

Comparison

	Default K8s	Volcano	Kueue	YuniKorn
Gang scheduling	No	Yes	Limited	Yes
Hierarchical queues	No	Yes	Yes	Yes
DRF fair-share	No	Yes	Basic	Yes
Custom scheduler	—	own	uses default	own
Best for	services	Spark·Flink·ML	queueing batch	Spark heritage

Demo

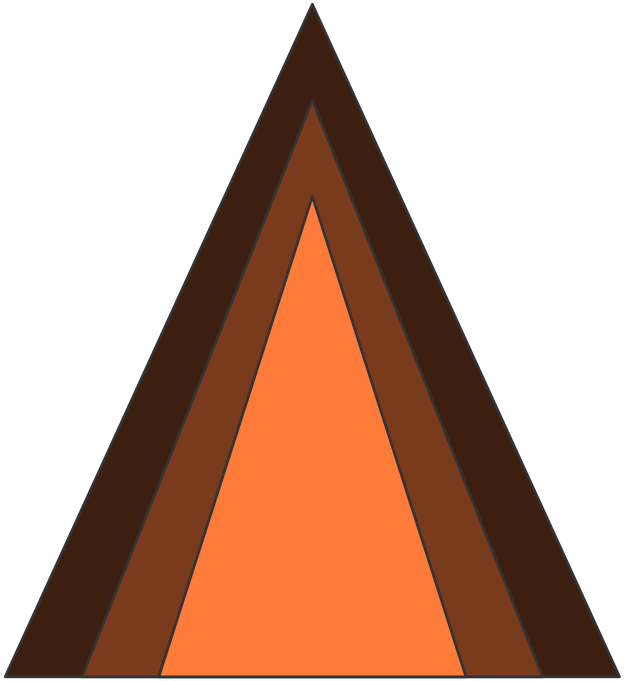
Spark-style gang scheduling — live

$ kind cluster · 3 workers · Volcano v1.10

A	daily-orders-rollup	3 executors · fits · all up
B	quarterly-revenue-rebuild	30Gi gang vs 23Gi cluster · all pending
C	quarterly-revenue-naive	same ask without Volcano · partial chaos

Thank You

Guilherme Nogueira



volcano.sh